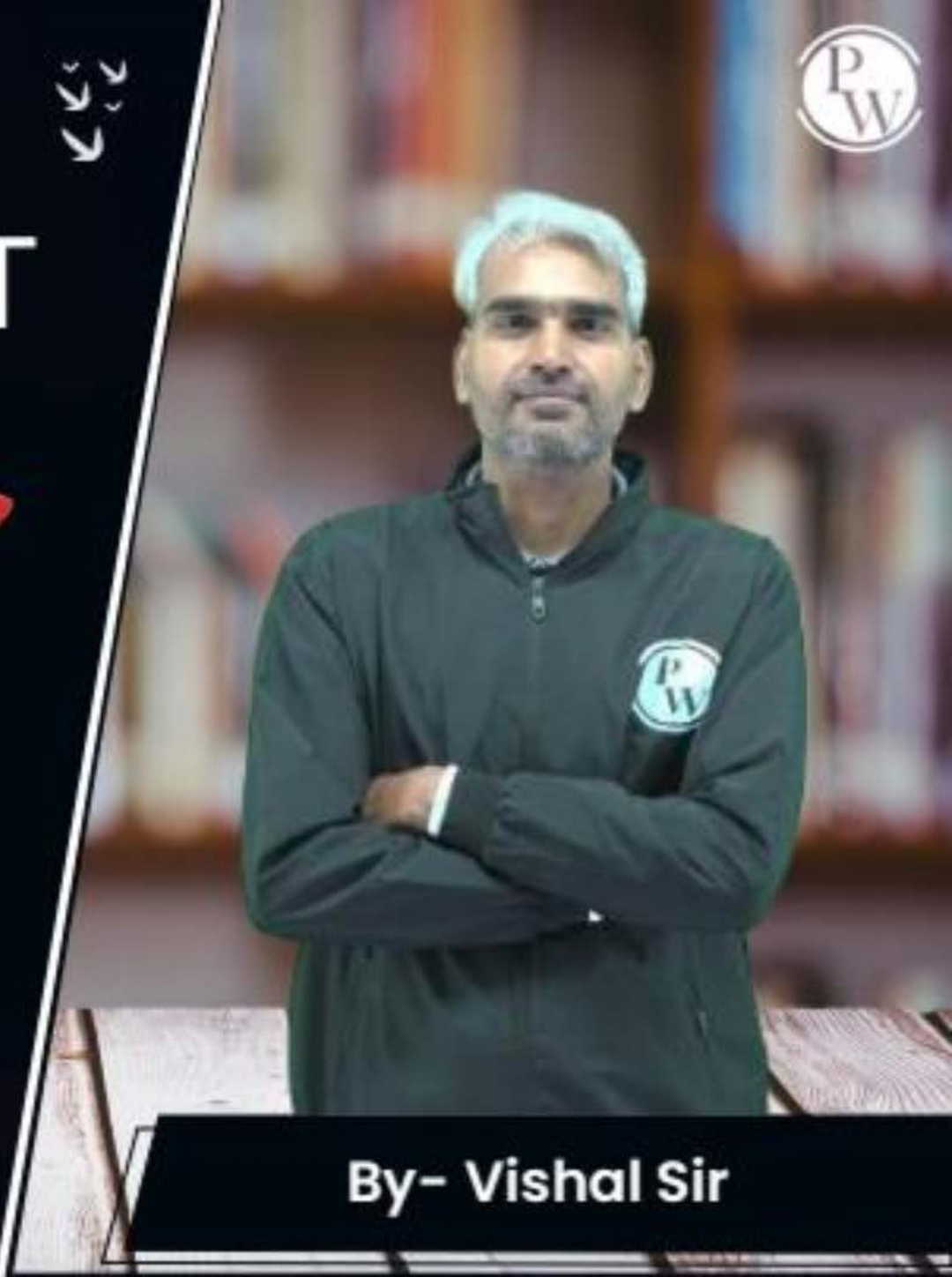
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



Topic Minimal cover or Canonical cover



Slide

Topics to be Covered



- Topic: Minimal cover or Canonical cover
- Topic: Schema refinement (Normalization)
- Topic: Problems because of redundancy in a relation
- Topic: Properties of decomposition

Slide

#e.g., Consider the following FD set

$F = \{A \rightarrow BC$

$CD \rightarrow E$

$E \rightarrow C$

$D \rightarrow AEH$

$ABH \rightarrow BD$

$DH \rightarrow BC$

$\}$

Find minimal cover of F.

	Simplify RHS.	Eliminate Extra attribute from LHS.	Eliminate Redundant FD
$A \to BC$	$A \to B$	$A \to B$	$A \to B$
	$A \to C$	$A \to C$	$A \to C$
$CD \to E$	$CD \to E$	$D \to E$ { C is extraneous when D is present }	Redundant
$E \to C$	$E \to C$	$E \to C$	$E \to C$
$D \to AEH$	$D \to A$	$D \to A$	$D \to A$
	$D \to E$	$D \to E$	$D \to E$
	$D \to H$	$D \to H$	$D \to H$
$ABH \to BD$	~~$ABH \to B$~~ trivial		
$DH \to BC$	$ABH \to D$	$AH \to D$ { B is extraneous }	$AH \to D$
	$DH \to B$	$D \to B$ } H is extraneous	Redundant
	$DH \to C$	$D \to C$ }	Redundant

Union

$A \to BC$

$E \to C$

$D \to AEH$

$AH \to D$



Topic : Schema refinement (Normalization)

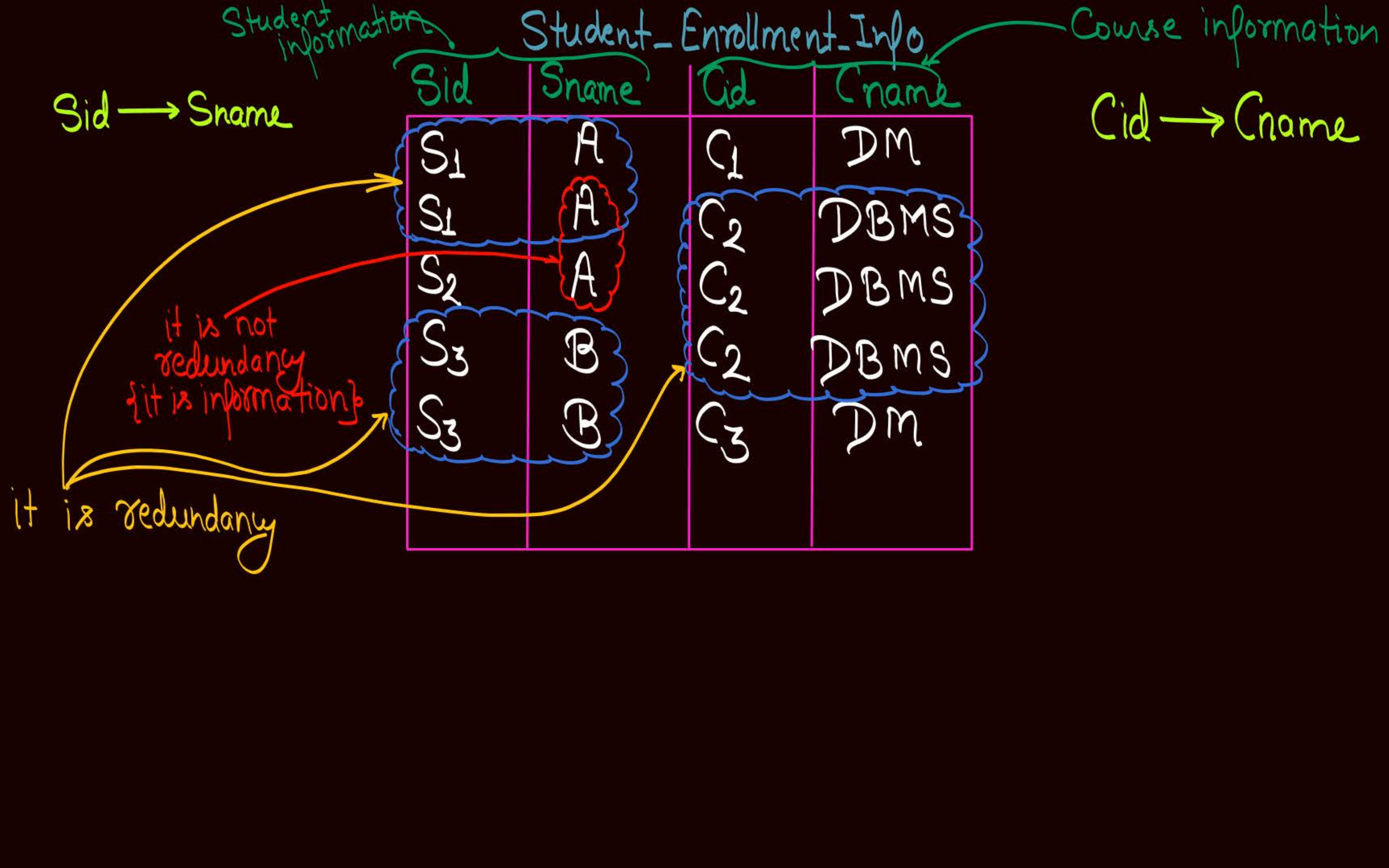
duplicate information

- Normalization is a process of reducing / eliminating the redundancy present in the relational table



Topic : Redundancy in a relation

If independent informations are stored in the same table, then redundancy is possible


Student information
Student_Enrollment_Info
Course information
Sid → Sname
Cid → Cname
Sid
Sname
Cid
Cname
S1
A
C1
DM
S1
A
C2
DBMS
S2
A
C2
DBMS
S3
B
C2
DBMS
S3
B
C3
DM
it is not redundancy {it is information}
it is redundancy

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

* Attributes of the primary key are denoted by "underlining" those attributes.

If {Sid, Cid} is the Primary key then Sid & Cid are not allowed to take "NULL" values

Let,

Sid $\longrightarrow$ Sname

&

Cid $\longrightarrow$ Cname

are only two FDs that holds in the relation.

∴ Candidate key of the relation is {Sid, Cid}

If we define a primary key for the relation then {Sid, Cid} will be Primary key

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Sid → Sname

Cid → Cname



Topic : Problems because of redundancy

* If redundancy is present in the relation, then various problems are possible

1. It requires more storage space { it is not a very big Problem }
2. Insertion Anomaly
3. Deletion Anomaly
4. Updation Anomaly

Insertion Anomaly :-

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM
NULL	NULL	C_4	AI

Sid → Sname

Cid → Cname

But Sid Can not be NULL

∴ We can not insert (C_4, AI) in the relation.

Insertion Anomaly:- If independent informations are stored in the same relational table, then sometimes it may not be possible to insert one information without inserting other independent information.

eg: We can not insert the information of a new Course (C4, AI) until some students enroll for that Course, because "Sid" can not be NULL.

Deletion Anomaly :-

Suppose we want to delete information of student with Sid : S_1

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1 NULL	A NULL	C_1	DM
S_1 NULL	A NULL	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Not allowed

∴ We will have to delete the Complete tuple

Sid → Sname

Cid → Cname

Deletion Anomaly :-

Suppose we want to delete information of student with Sid : S_1

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Sid → Sname

Cid → Cname

If we delete all the tuples w.r.t student "S_1", then we will lose the information of Course "C_1" as well.

Deletion Anomaly :- Some times when we try to delete one independent information, we may lose some other independent information.

eg: If we try to delete the information of student "S_1" then information of Course with $Cid = C_1$ is lost.

Updation Anomaly :-

↓

Let us consider a situation in which we want to update the name of the Course with Cid = C_2 to DB & WH from DBMS

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	~~DBMS~~ DB & WH
S_2	A	C_2	~~DBMS~~ DB & WH
S_3	B	C_2	~~DBMS~~ DB & WH
S_3	B	C_3	DM

Sid → Sname

Cid → Cname

↳ Updation will be required in all duplicate Copies ⟹ ∴ This operation will be time Consuming

Updation Anomaly :-

Updation is required in all duplicate Copies, so it will be time consuming operation



Topic : Schema refinement (Normalization)

- Normalization is a process of decomposing (splitting) a relational tables into smaller tables (sub-relations) such that it eliminates/reduces the data redundancy, and it can overcome undesirable characteristics like Insertion, Updation and Deletion Anomalies

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

$F = \{ Sid \rightarrow Sname,\ Cid \rightarrow Cname \}$

Assume that the relation is decomposed into following three sub-relations

Student

Sid	Sname
S_1	A
S_2	A
S_3	B

$F_1 = \{Sid \rightarrow Sname\}$

C.K = {Sid}

Course

Cid	Cname
C_1	DM
C_2	DBMS
C_3	DM
C_4	AI

$F_2 = \{Cid \rightarrow Cname\}$

C.K = {Cid}

Enroll

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_3	C_2
S_3	C_3

$F_3 = \{$ Empty $\}$

C.K = {Sid, Cid}

After decomposition there is no redundancy & After decomposition all anomalies are overcome.



2 mins Summary

- Topic: Minimal cover or Canonical cover
- Topic: Schema refinement (Normalization)
- Topic: Problems because of redundancy in a relation
- Topic: Properties of decomposition

Slide

THANK - YOU